EXPERIMENT NO: 10

File Handling and Counting Character Patterns

1. Basic Information

Program Name: Experiment10.java

Package: recordprog

Language: Java

2. Objective / Aim

To develop a Java program that writes a given passage to a text file, reads the contents back from the file, and counts the occurrences of the character patterns “pe” and “pi”, without distinguishing between uppercase and lowercase letters.

3. Description

The program stores a tongue-twister passage in a string and writes it to **sample.txt** using FileWriter. It then reads the file character by character using FileReader and stores the contents in a string. The content is converted to lowercase so the search is case-insensitive. A loop checks each pair of adjacent characters and counts occurrences of “pe” and “pi”. The counts are displayed on the screen. IOException is caught if a file operation fails.

4. System Requirements

Hardware: Standard PC/Laptop; minimum 4 GB RAM; minimum 500 MB free storage.

Software: Windows/Linux/macOS; Java Development Kit (JDK); IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code; Command Prompt/Terminal.

5. Algorithm

1. Start the program.

2. Store the given tongue-twister passage in a string.

3. Create a FileWriter and write the string into sample.txt.

4. Close the writer.

5. Open sample.txt using FileReader and read characters until the end of the file.

6. Close the reader and convert the content to lowercase.

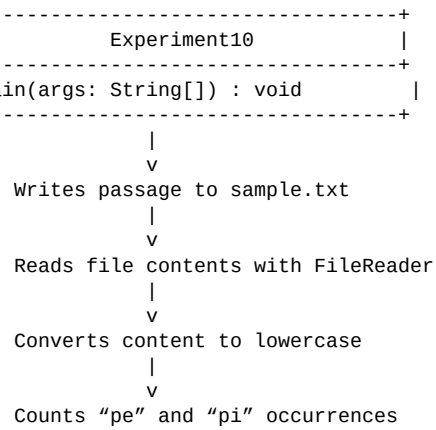
7. Initialize the counters for “pe” and “pi” to zero.

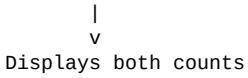
8. Check each pair of adjacent characters in the content.

9. Increase the corresponding counter when “pe” or “pi” is found.

10. Display both counts and handle any IOException.

6. UML / Program Structure




Displays both counts

7. Flow of Execution

```
Start
  |
  v
Prepare passage text
  |
  v
Write text to sample.txt
  |
  v
Read file contents
  |
  v
Convert content to lowercase
  |
  v
Initialize peCount = 0, piCount = 0
  |
  v
Check each pair of adjacent characters
  |
  v
Is pair “pe”? ---- Yes ---> Increment peCount
  |
  v
Is pair “pi”? ---- Yes ---> Increment piCount
  |
  v
Display both counts
  |
  v
Stop
```

8. Source Code

```
package recordprog;
import java.io.FileWriter;
import java.io.FileReader;
import java.io.IOException;

public class Experiment10 {
    public static void main(String[] args) {
        String text =
            "Peter Piper picked a peck of pickled peppers\n" +
            "A peck of pickled peppers Peter Piper picked\n" +
            "If Peter Piper picked a peck of pickled peppers\n" +
            "Where’s the peck of pickled peppers Peter Piper picked?";
        try {
            // Writing data into the file
            FileWriter writer = new FileWriter("sample.txt");
            writer.write(text);
            writer.close();

            // Reading data from the file
            FileReader reader = new FileReader("sample.txt");
            String content = "";
            int ch;
            while ((ch = reader.read()) != -1) {
                content += (char) ch;
            }
            reader.close();

            // Convert to lowercase for counting
            content = content.toLowerCase();
            int peCount = 0;
            int piCount = 0;

            // Count "pe" and "pi"
            for (int i = 0; i < content.length() - 1; i++) {
                if (content.substring(i, i + 2).equals("pe")) {
                    peCount++;
                }
                if (content.substring(i, i + 2).equals("pi")) {
                    piCount++;
                }
            }
            System.out.println("'pe' - no of occurrences - " + peCount);
            System.out.println("'pi' - no of occurrences - " + piCount);
```

```
        } catch (IOException e) {
            System.out.println("An error occurred.");
        }
    }
}
```

9. Compilation and Execution

```
javac -d . Experiment10.java\njava recordprog.Experiment10
```

10. Sample Output

Example:

```
'pe' - no of occurrences - 13
'pi' - no of occurrences - 6
```

11. Result

Thus, the Java program to write text to a file, read the contents, and count the occurrences of “pe” and “pi” was written and executed successfully.